

WELL COMPLETION REPORT (SYNTHETIC)

OIL-KMC-005 - Kumchai-005

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000060
Field / District	Kumchai / Tinsukia
Block	OALP-IV-04
Basin	Assam-Arakan
Formation	Lakadong+Therria (LKT)
Spud / Completion	2016-12-27 / 2017-03-05
Measured / TVD	3735 m / 3715 m
Trajectory	Vertical; inc 4.7 deg; az 309.7 deg
Formation Interval	2194-3403 m
Lithology	Sandstone with carbonaceous interbeds
Pore Pressure Reference	4224 psi
Fracture Gradient	14.60 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Moderate (53.2/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	933	Surface
9 5/8 in casing	2166	Intermediate
7 in liner/casing	3286	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2017-01-31 2050 m	Torque Spike	Critical	Rotary speed fluctuated during constant pump rate. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2017-01-29 1949 m	Fishing Operation	High	Downhole tool remained unrecovered after freeing attempts. Mitigation: Confirm fish geometry before selecting retrieval tool. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2017-01-29 1974 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2017-02-01 2100 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Avoid prolonged static periods in high-drag interval.
2017-01-22 1617 m	Mud Loss	Medium	Flow-out dropped below flow-in with pit loss. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2017-02-24 3255 m	Mud Loss	Medium	Active pit volume declined during stable pump rate. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2017-01-05 686 m	Cementing Issue	Low	Cement density/returns trend indicated placement uncertainty. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Offset loss history should inform cementing hydraulics.

Completion Summary

The synthetic well reached 3735 m MD in the Lakadong+Therria interval. The NWIS knowledge record links 8 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.